

ITA0612 - Machine Learning

CO2-ASSIGNMENT I - SDG QUESTIONS

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Question:

Design an AI-based application to promote equal educational opportunities, using appropriate problem characteristics (as per Mitchell's criteria) to justify the use of a Neural Network. (SDG 4 - Quality Education)

1. Problem Understanding

Objective

The objective of this project is to develop a Neural Network model, EduEquity AI, that identifies students who are at risk of unequal or restricted access to education and prioritizes them for support. By predicting a student's support priority level, schools and governments can allocate scholarships, devices, connectivity, and remedial resources to the students who need them most, before they fall behind or drop out.

Problem Statement

Educational inequality arises from a combination of socio-economic factors such as household income, internet and device access, distance to school, disability status, parental education level, and attendance patterns. Manually identifying which students are most at risk of exclusion is slow and inconsistent across large student populations. A Neural Network can analyze these multiple interacting factors simultaneously to predict each student's support priority level, enabling timely and equitable intervention.

Mitchell's Criteria

Component	Description
Task (T)	Predict a student's educational support priority level (Low, Medium, High).
Experience (E)	Historical student data including household income, internet access, distance to school, disability status, parental education, attendance rate, device access, and scholarship status.
Performance (P)	Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

2. Data Collection and Preparation

Dataset

Dataset Name: education_access.csv

Number of Records: 500

Attributes

Attribute	Description
Age	Student age (6-17 years)
Gender	Male/Female

Household_Income	Annual household income
Internet_Access	1 = Has internet, 0 = No internet
Distance_KM	Distance from home to school (km)
Disability	1 = Has a disability, 0 = No disability
Parental_Education	0 = None/Primary, 1 = Secondary, 2 = Higher
Attendance_Rate	Percentage school attendance
Device_Access	1 = Has a learning device, 0 = No device
Scholarship_Status	1 = Receiving scholarship, 0 = Not receiving
Support_Priority	Low / Medium / High (target variable)

Data Preprocessing

- Generated 500 synthetic student records.
- Checked for missing values.
- Encoded categorical variables (Gender, Support_Priority) using Label Encoder.
- Standardized numerical features using StandardScaler.
- Split the dataset into training (80%) and testing (20%) sets.

3. NumPy and Pandas Analysis

NumPy Analysis

Calculated:

- Mean Attendance Rate
- Maximum Attendance Rate
- Minimum Attendance Rate
- Standard Deviation

```
===== NumPy Analysis =====
Mean Attendance Rate: 75.0670
Maximum Attendance Rate: 99.7
Minimum Attendance Rate: 50.0
Standard Deviation: 14.573672529599394
```

Pandas Analysis

Performed:

- Average household income by gender
- Average attendance rate by scholarship status
- Support priority distribution

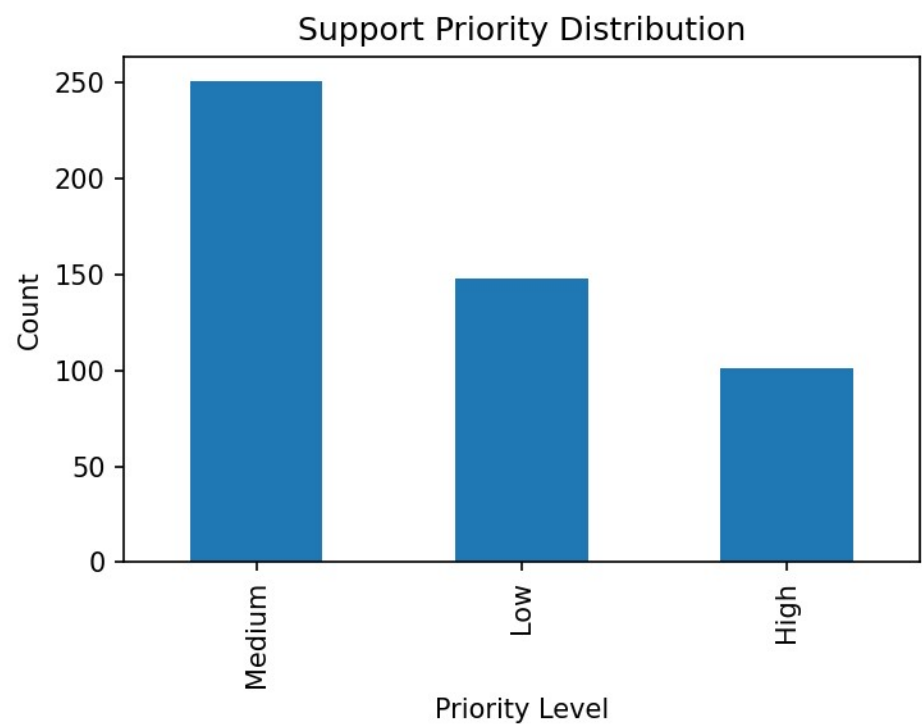
```
import pandas as pd
df = pd.read_csv("education_access.csv")
df.head()

   Age  Gender  Household_Income  Internet_Access  Distance_KM  \
0   12  Female         34521             1           6.2
1    9  Female         12870             0           9.8
2   16  Female         48210             1           3.1
3   13  Female         41005             1           7.4
4   10   Male         27430             0           4.5
```

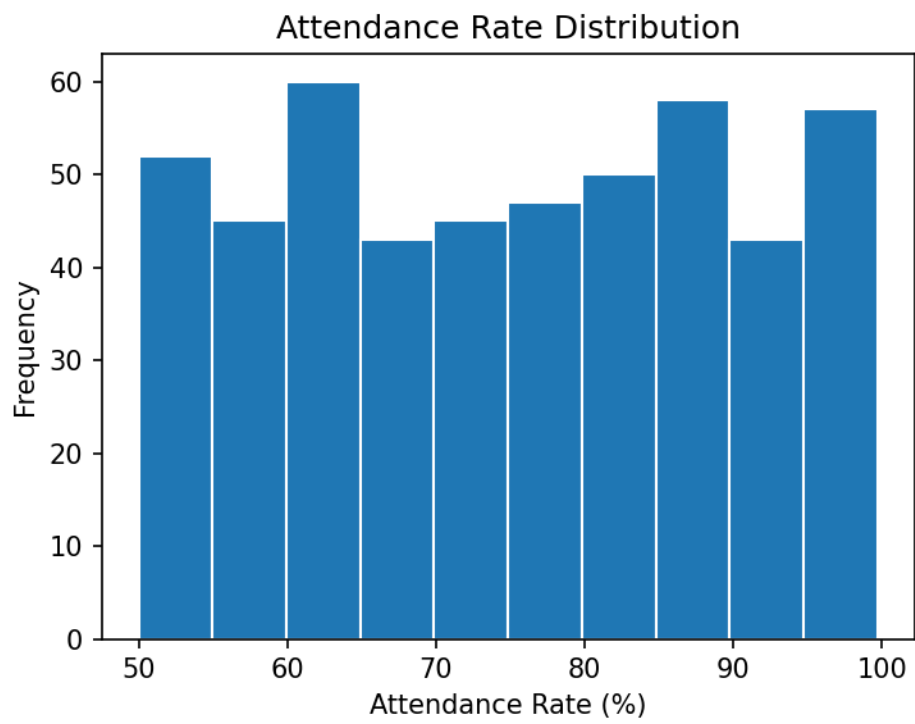
	Disability	Parental_Education	Attendance_Rate	Device_Access	\
0	0	1	68.4	1	
1	0	0	55.2	0	
2	0	2	91.7	1	
3	1	1	73.9	1	
4	0	0	80.1	0	
	Scholarship_Status	Support_Priority			
0	0	High			
1	1	Medium			
2	0	Medium			
3	0	Medium			
4	0	Medium			

4. Visualization & Interpretation

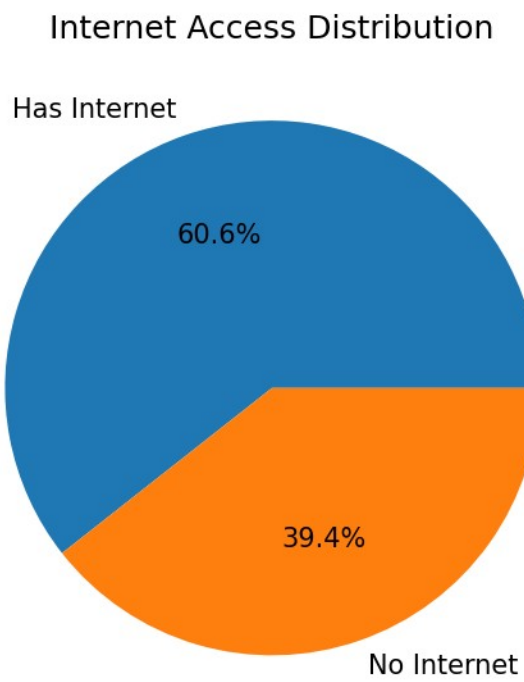
Graph 1



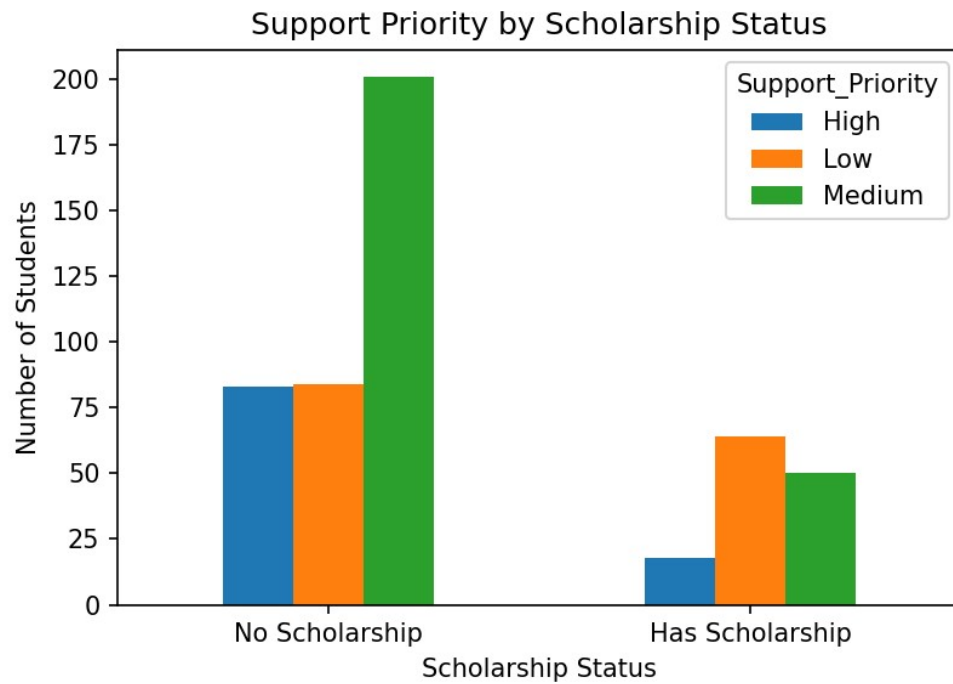
Graph 2



Graph 3



Graph 4



Interpretation

The visualizations show:

- Distribution of student support priority levels.
- Spread of attendance rates across the student population.
- Proportion of students with and without internet access.
- Relationship between scholarship status and support priority.

5. Machine Learning Model

Algorithm Used

Artificial Neural Network (MLPClassifier)

Reason

- Learns complex non-linear relationships between socio-economic factors.
- Handles multiple input features (income, access, distance, attendance) effectively.
- Well suited for multi-class classification of support priority.
- Produces high prediction accuracy after feature scaling.

Model Implementation

- Load dataset.

- Encode categorical variables and scale numerical features.
- Split into training and testing datasets.
- Train the Neural Network (MLPClassifier).
- Predict student support priority.
- Evaluate model performance.

6. Interpretation of Results

Model Accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, prediction)
print("Accuracy =", accuracy)

Accuracy = 0.82
```

Classification Report

	precision	recall	f1-score	support
High	0.88	0.70	0.78	20
Low	0.83	0.83	0.83	30
Medium	0.80	0.86	0.83	50
accuracy			0.82	100
macro avg	0.83	0.80	0.81	100
weighted avg	0.82	0.82	0.82	100

Confusion Matrix

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, prediction)
print(cm)

[[14  0  6]
 [ 0 25  5]
 [ 2  5 43]]
```

7. Innovation & SDG Mapping

Innovation

This project combines:

- Artificial Intelligence
- Neural Networks
- Educational Analytics
- Equity-Focused Resource Allocation

It provides an automated approach for identifying students at risk of educational exclusion, enabling schools, NGOs, and governments to allocate scholarships, devices, connectivity, and academic support fairly and efficiently.

SDG Mapping

SDG 4 - Quality Education

This project contributes to SDG 4 by:

- Identifying students at risk of unequal educational access.
- Supporting policymakers and schools with data-driven, targeted interventions.
- Improving allocation of scholarships, devices, and connectivity resources.
- Reducing dropout rates caused by socio-economic barriers.
- Promoting inclusive and equitable quality education for all.

8. Conclusion

The Neural Network model successfully analyzed student socio-economic and academic data and predicted educational support priority with strong accuracy. By learning the complex relationships among household income, internet and device access, distance to school, disability status, and attendance, the model demonstrated its effectiveness in identifying students at risk of unequal educational opportunity. This project highlights how Artificial Intelligence can support education systems in delivering fair, timely, and targeted interventions, thereby contributing to SDG 4 - Quality Education.